

OptiCrop: Smart Agricultural Production Optimization Engine

Project Description

OptiCrop is a Machine Learning-based web application that recommends the most suitable crop based on soil nutrients and environmental conditions. The system analyzes parameters such as Nitrogen (N), Phosphorous (P), Potassium (K), temperature, humidity, pH, and rainfall to help farmers make informed crop selection decisions, improve productivity, and support sustainable agriculture.

Technical Architecture

User Input → Flask Application → Data Preprocessing → Machine Learning Model → Crop Prediction Engine → Recommended Crop Result

Milestone 1: Dataset Preparation and Model Selection

Load the Crop_recommendation.csv dataset, analyze the data, perform data preprocessing, handle missing values (if any), and prepare the dataset for Machine Learning model training.

Milestone 2: Model Development

Train and evaluate the Machine Learning model using the crop recommendation dataset. Save the trained model as model.pkl along with scaler.pkl and label_encoder.pkl for prediction.

Milestone 3: Flask Backend Development

Develop app.py to load the trained model, process user inputs, perform crop prediction, and return the recommended crop through Flask routes.

Milestone 4: Frontend Development

Design responsive web pages using HTML and CSS (index.html, findyourcrop.html, about.html, and result.html) to collect user input and display crop recommendations..

Milestone 5: Deployment

Install the required dependencies from requirements.txt, run the Flask application locally, and access the system through a web browser to generate crop recommendations.

Conclusion

The OptiCrop project demonstrates the practical application of Machine Learning in smart agriculture by providing accurate crop recommendations based on soil and environmental conditions. It helps farmers improve crop selection, increase agricultural productivity, and promote efficient resource utilization.